

Web Application Development Practical - Angular Application

Practical: Create an Angular application which will perform Register User, Login User and Show User Data on Profile Component.

Aim: To develop an Angular application with user registration, login and profile display functionality.

Requirements:

1. Node.js Installed
2. Angular CLI Installed
3. VS Code Editor
4. Internet Browser

Step 1: Install Angular CLI using `npm install -g @angular/cli`.

Step 2: Create Angular project using `ng new user-app`.

Step 3: Open project in VS Code.

Step 4: Create components using Angular CLI.

Step 5: Create Register Component.

Step 6: Create Login Component.

Step 7: Create Profile Component.

Step 8: Configure Routing Module.

Step 9: Add FormsModule in `app.module.ts`.

Step 10: Run project using `ng serve` command.

Step 11: Open browser and visit `http://localhost:4200`.

Commands

```
ng new user-app
cd user-app
ng generate component register
ng generate component login
ng generate component profile
ng serve
```

app.module.ts

```
import { NgModule } from '@angular/core';
import { BrowserModule } from '@angular/platform-browser';
import { FormsModule } from '@angular/forms';

import { AppComponent } from './app.component';
import { RegisterComponent } from './register/register.component';
import { LoginComponent } from './login/login.component';
import { ProfileComponent } from './profile/profile.component';
@NgModule({
```

```

    declarations: [
      AppComponent,
      RegisterComponent,
      LoginComponent,
      ProfileComponent
    ],

    imports: [
      BrowserModule,
      FormsModule
    ],

    providers: [],
    bootstrap: [AppComponent]
  })

  export class AppModule { }

```

register.component.html

```

<h2>User Registration</h2>

<form>

  <input type="text" placeholder="Enter Name" [(ngModel)]="name" name="name">
  <br><br>

  <input type="email" placeholder="Enter Email" [(ngModel)]="email" name="email">
  <br><br>

  <input type="password" placeholder="Enter Password" [(ngModel)]="password" name="password">
  <br><br>

  <button (click)="registerUser()">Register</button>

</form>

```

register.component.ts

```

import { Component } from '@angular/core';

@Component({
  selector: 'app-register',
  templateUrl: './register.component.html'
})

export class RegisterComponent {

  name = '';
  email = '';
  password = '';

  registerUser() {

    localStorage.setItem('name', this.name);
    localStorage.setItem('email', this.email);
    localStorage.setItem('password', this.password);

    alert('User Registered Successfully');
  }
}

```

login.component.html

```

<h2>User Login</h2>

<form>

  <input type="email" placeholder="Enter Email" [(ngModel)]="email" name="email">
  <br><br>

  <input type="password" placeholder="Enter Password" [(ngModel)]="password" name="password">
  <br><br>

```

```

        <button (click)="loginUser()">Login</button>
    </form>

```

login.component.ts

```

import { Component } from '@angular/core';

@Component({
  selector: 'app-login',
  templateUrl: './login.component.html'
})

export class LoginComponent {

  email = '';
  password = '';

  loginUser() {

    let storedEmail = localStorage.getItem('email');
    let storedPassword = localStorage.getItem('password');

    if(this.email == storedEmail && this.password == storedPassword){

      alert('Login Successful');

    } else {

      alert('Invalid Email or Password');

    }
  }
}

```

profile.component.ts

```

import { Component } from '@angular/core';

@Component({
  selector: 'app-profile',
  templateUrl: './profile.component.html'
})

export class ProfileComponent {

  name = localStorage.getItem('name');
  email = localStorage.getItem('email');

}

```

profile.component.html

```

<h2>User Profile</h2>

<p><b>Name:</b> {{name}}</p>

<p><b>Email:</b> {{email}}</p>

```

Explanation:

Angular is a front-end framework used to build single page applications. In this application, Register Component stores user details using localStorage. Login Component verifies user credentials and Profile Component displays user information. FormsModule is used for two-way data binding using ngModel.

Result: Angular application for user registration, login and profile display was created successfully.